📖 Read: Atomic Design & Building with Systems

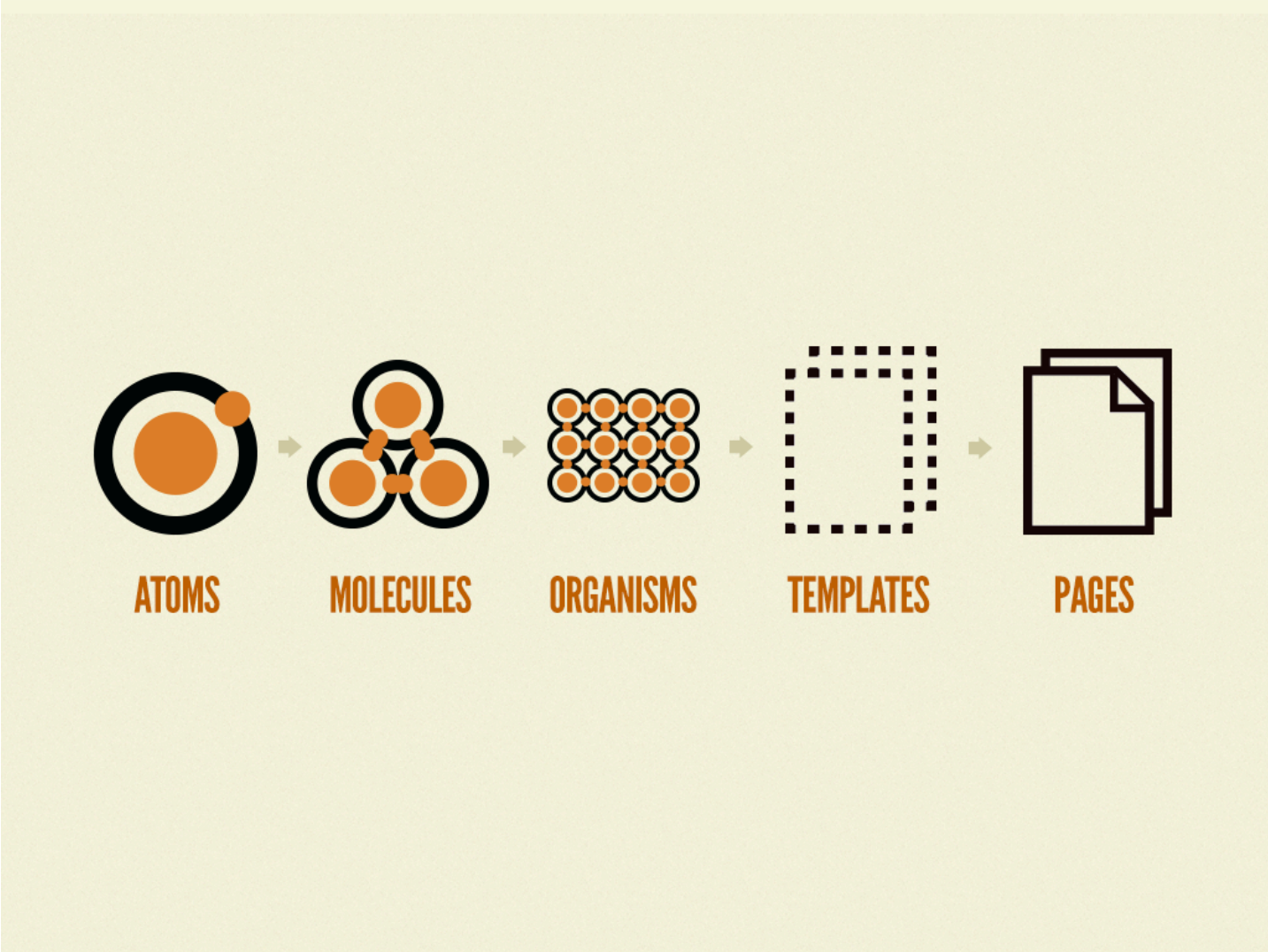
Read: Atomic Design & Building with Systems

As a new designer, it is tempting to design screen by screen, treating each as an individual work of art. The professional designer, however, thinks in systems. They follow the principle of Atomic Design, a methodology introduced by designer Brad Frost that has become an industry standard. The core idea is that interfaces are not monolithic pages; they are collections of reusable components.

> *"We're not designing pages, we're designing systems of components."*
>
> —Brad Frost, Atomic Design

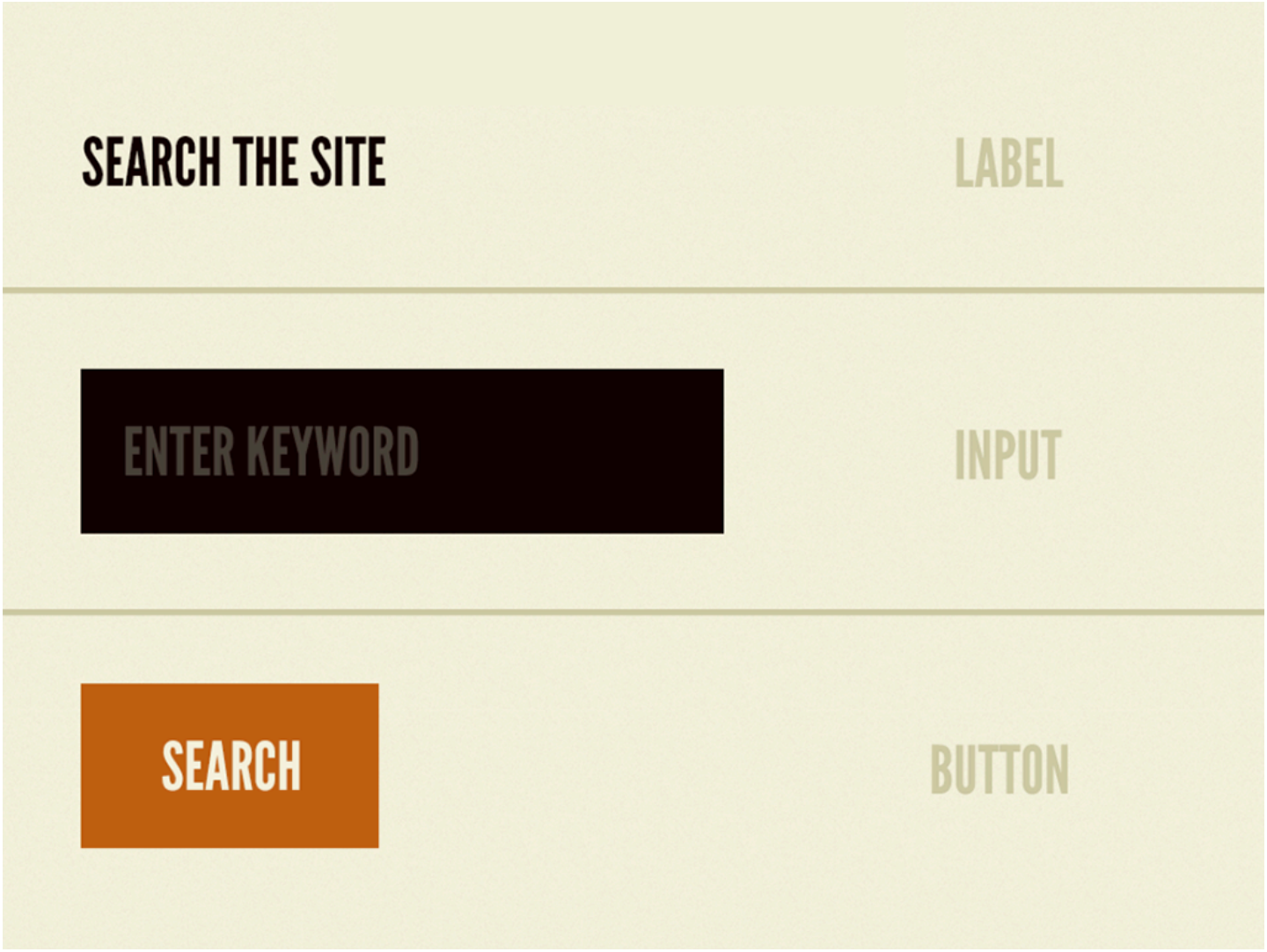
The Hierarchy of a System

Frost provides a simple analogy from chemistry to help us understand this system.



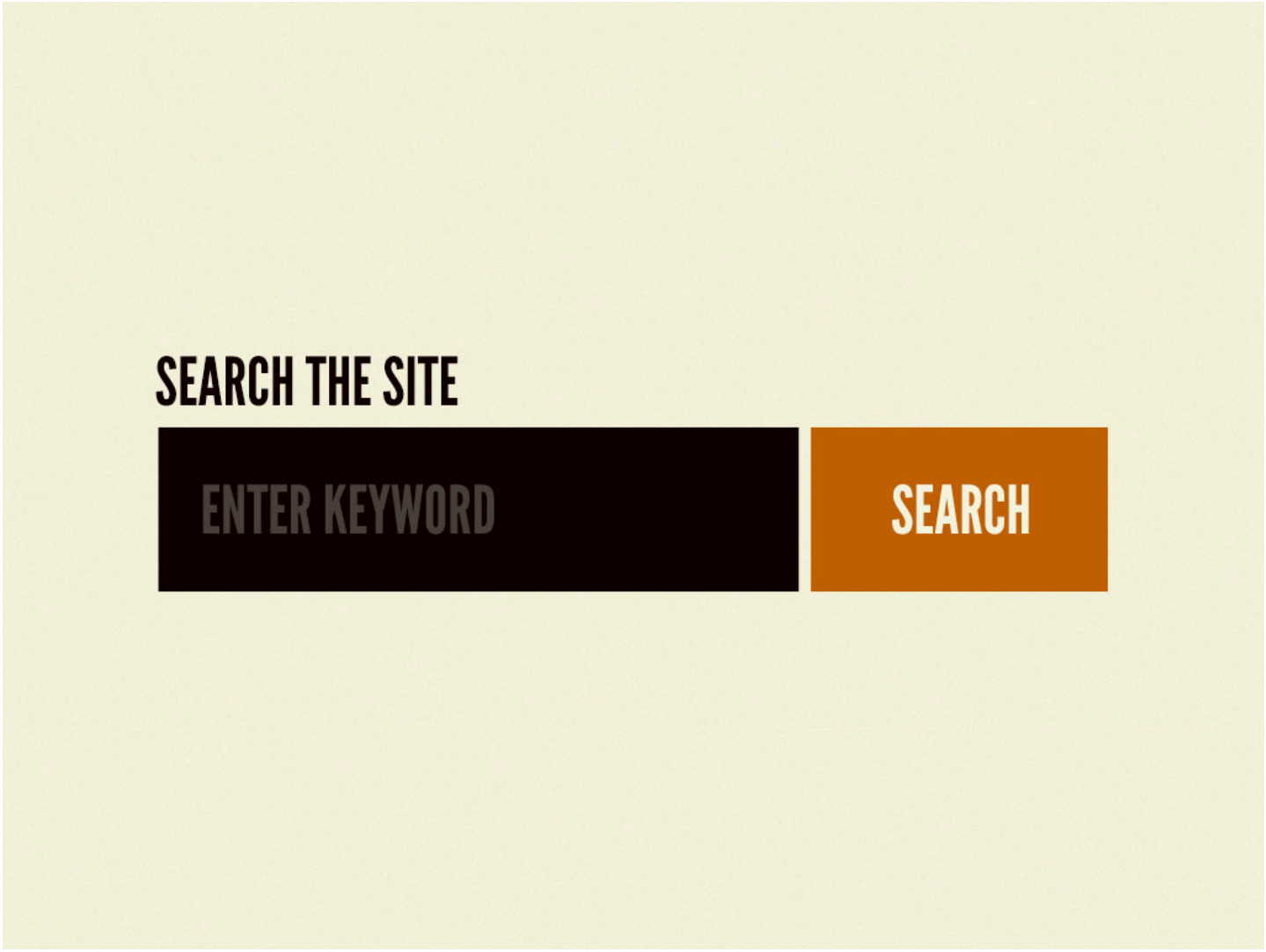
Atoms:

These are the indivisible UI elements. They are the absolute basics, such as a text label, an input field, or a button. On their own, they are not very useful, but they are the foundational building blocks.



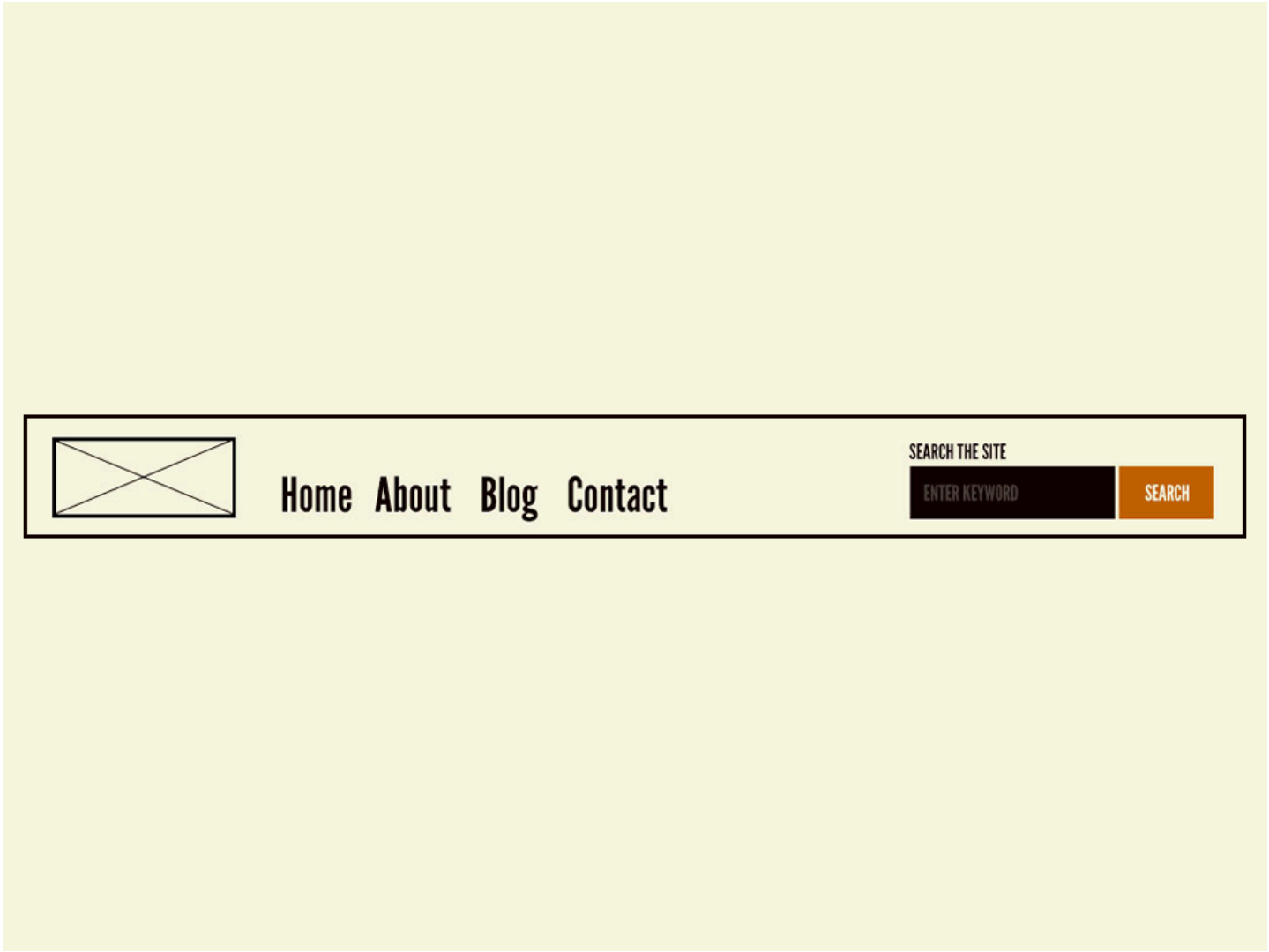
Molecules:

These are simple groups of atoms that function together as a unit. For example, a search form molecule might be composed of a label atom, an input field atom, and a button atom.



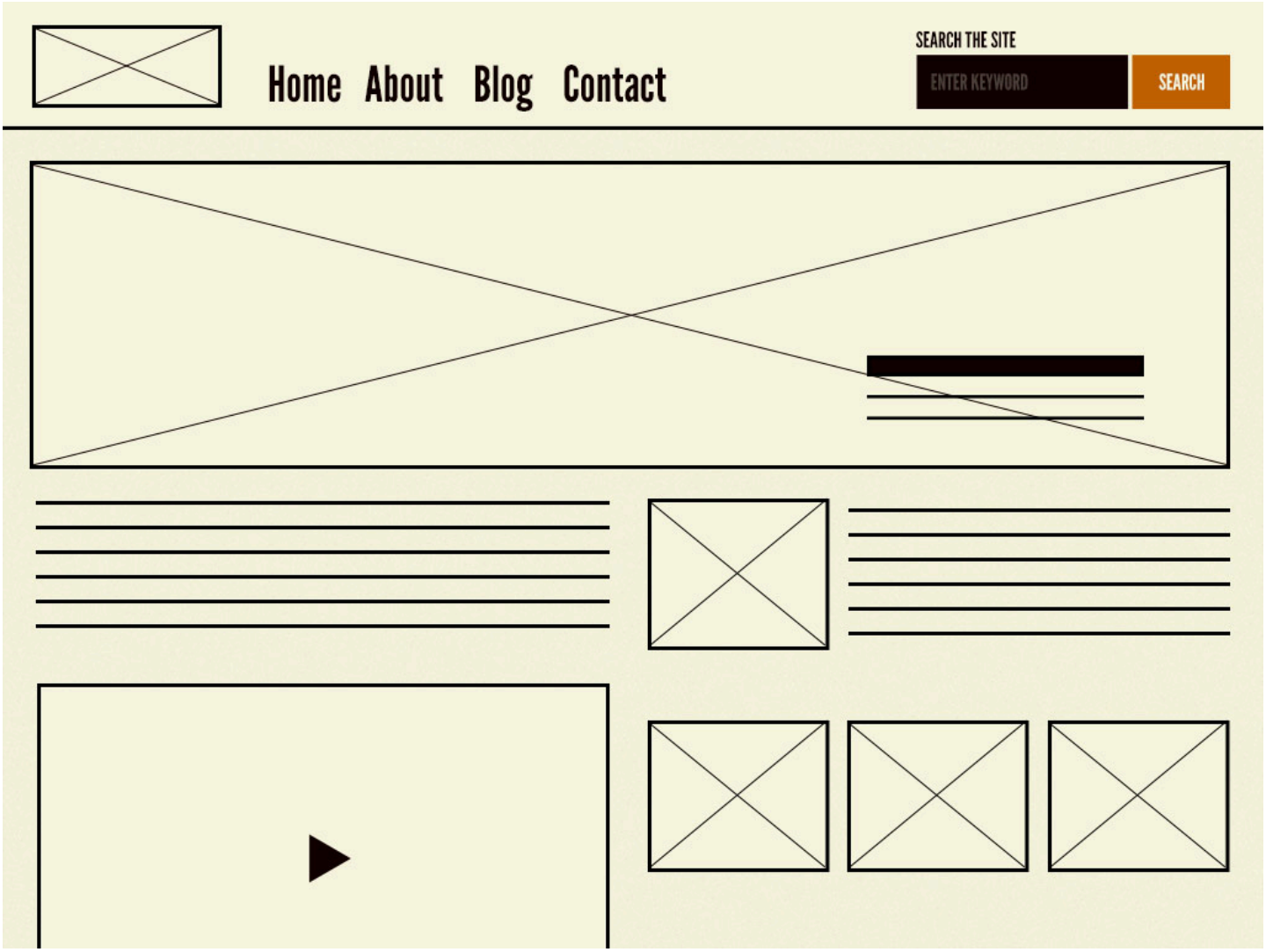
Organisms:

These are more complex components that form distinct sections of an interface. An organism might be a site header, composed of a logo molecule, a navigation molecule, and a search form molecule.



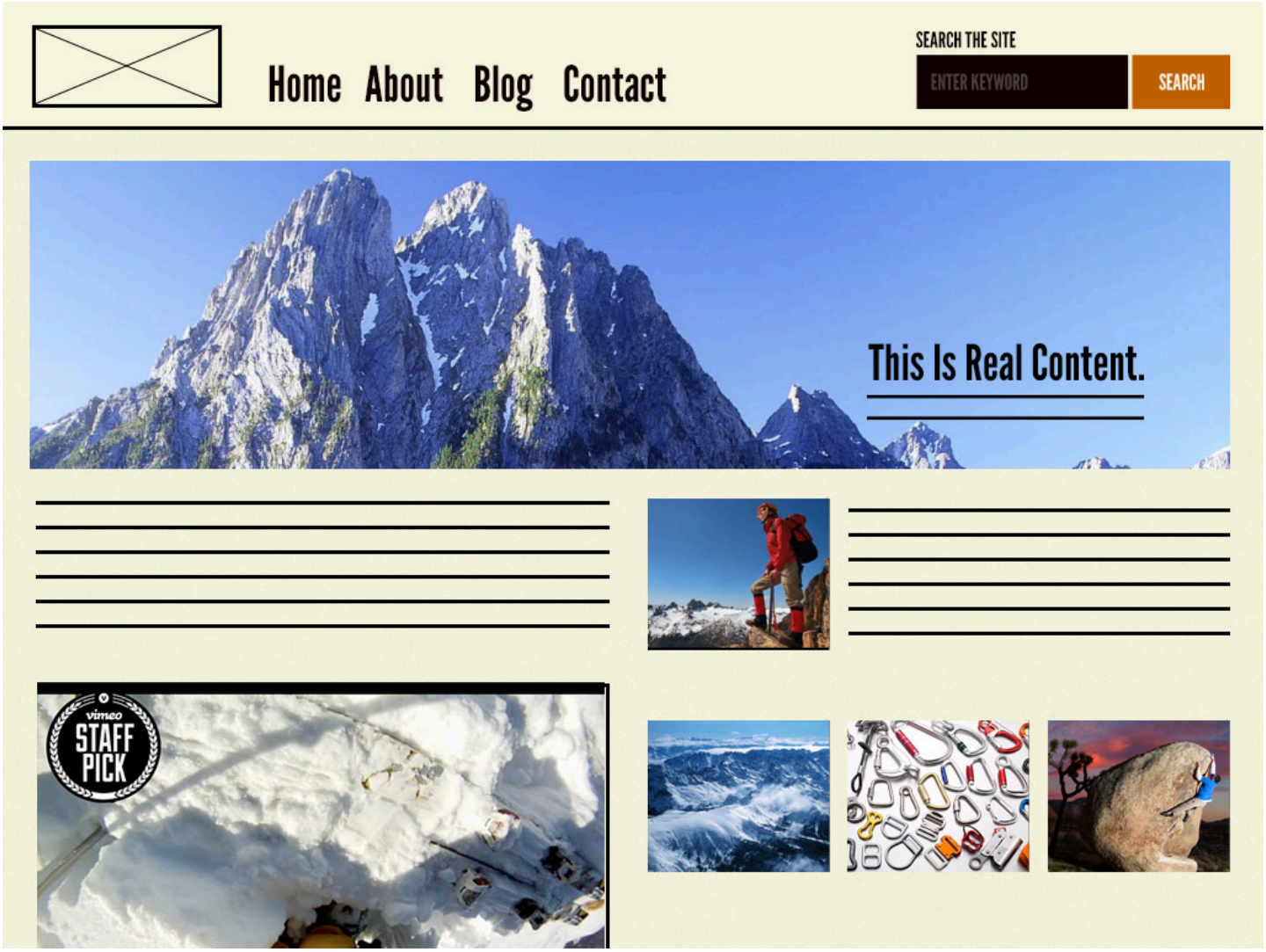
Templates:

These are page-level structures that place components into a layout. Templates show how organisms work together to form complete interfaces. They use placeholder content to demonstrate structure.



Pages:

These are specific instances of templates filled with real content. Pages show what users actually see. They help test how the design system works with actual data.

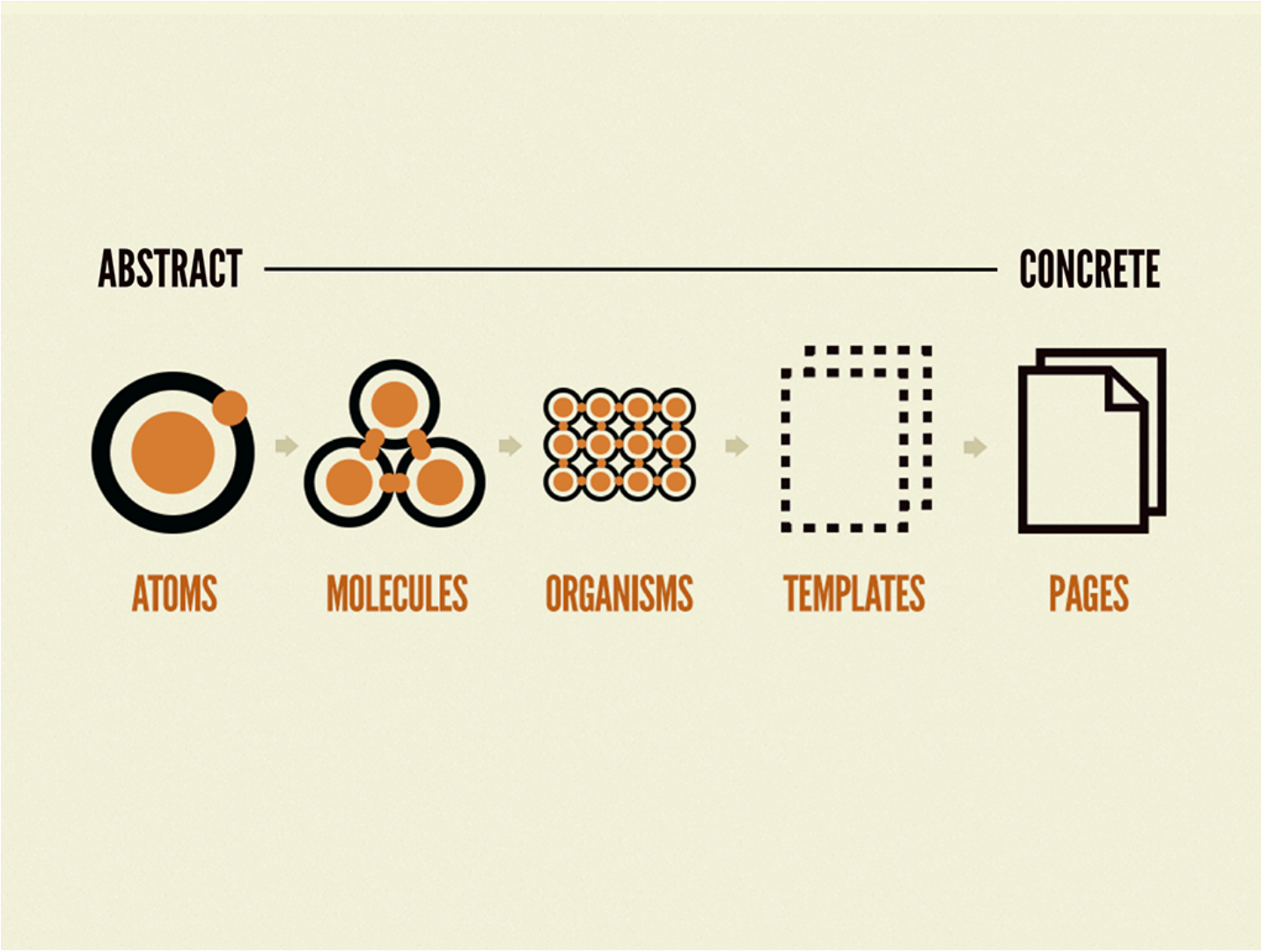


Moving Between Abstract and Concrete While Designing

By building from small, simple atoms up to complex organisms, you create a design system that is both consistent and scalable. Additionally, the structure and flexibility of

a design system, or at least a well-built one, allows you to move around in a nonlinear manner. What does that mean? Simply that you, as a product designer, can move between the abstract and concrete parts of the system.

This flexibility provides product designers with a constant reminder that the user interfaces in our digital products are built from smaller bits and pieces. While most users never consider this fact, designers building digital products always do. Constantly.



> *The painter, when at a distance from the easel, can assess and*

> *analyze the whole of the work from this vantage. He scrutinizes and listens, chooses the next stroke to make, then approaches the canvas to do it. Then, he steps back again to see what he's done in relation to the whole. It is a dance of switching contexts, a pitter-patter pacing across the studio floor that produces a tight feedback loop between mark-making and mark-assessing.*
>
> —Frank Chimero, The Shape of Design

Understanding Design System Terminology

So, what does this mean for you, practically? Why is *Atomic Design* important for digital product designers? First, it is a very helpful source for initially learning about design systems, which is extremely important for new designers to understand. Systems and product design are linked, after all.

And, again, remember that terminology surrounding design systems differs. The design systems that most digital product teams use in 2025 do not use most of the terminology from Atomic Design. Yes, terms like "atoms," "molecules," and "organisms" appear in some design conversations. However, professional teams more commonly use simpler language, such as "components," "primitives," or "patterns," or simply describe things by their function, such as "button component," "card pattern," or "navigation bar."

What matters is understanding the underlying principle: Break complex interfaces into smaller, reusable pieces that work together as a system. The chemistry metaphor from Atomic Design helps you learn this concept. The specific names matter less than the systematic thinking. Think about the context. Think about the full picture. Think and design in systems.

> *The names I've chosen for the stages of atomic design have worked really well for me and the teams I've worked with as we create UI design systems. But maybe they don't work for you and your organization.*
>
> *...*
>
> *"Atomic design" as a buzzword encapsulates the concepts of modular design and development, which becomes a useful shorthand for convincing stakeholders and talking with colleagues. But **atomic design is not rigid dogma**. Ultimately, whatever taxonomy you choose to work with should help you and your organization communicate more effectively in order to craft an amazing UI design system.*
>
> —Brad Frost, Atomic Design

Building Your Own Design System

For Project 02, you will create a simplified design system for your digital product. Professional design systems can be massive, comprising hundreds of components, detailed documentation, and complex guidelines. Your design system will be much smaller and focused.

Your design system should include:

- **Color Palette:** Primary colors, secondary colors, and neutral colors (grays, blacks, whites)

- **Typography System:** Font choices, sizes, and weights for headings, body text, and labels
- **Spacing Values:** Consistent measurements for margins and padding
- **Key Components:** Buttons, cards, forms, navigation, and other repeated elements

Professional designers sometimes refer to these simplified design systems as "style guides" or "design documentation." The name may vary, but the purpose remains the same. You are documenting your visual design decisions so you can apply them consistently.

From Theory to Practice in Figma

This approach has a direct parallel in Figma. By creating a master component for each repeated element, you can reuse it throughout your designs. If you need to change the color of every primary button in your product, you only need to edit the master component. Every instance updates automatically.

Using Figma's auto-layout feature, you can create components that resize and adapt automatically. Using Figma's variants feature, you can build different states, such as hover, focused, or disabled, into a single, powerful component. This systematic approach ensures consistency, saves an enormous amount of time, and is the key to building digital products that feel coherent and professional.

You learned Figma's tools in Week 02. You learned visual design principles in Week 04. Now you will combine these skills with systematic thinking to create high-fidelity mockups.

▶ References